

Incident: Blast Chiller Temperature Excursion

1) Document Owner and Scope

- Owner: Samuel Okafor (Sous Chef, Prep)
- Station: Prep
- Applies to: Any cooked, TCS prep item entering the blast chiller; this incident focuses on a chili batch.

2) Incident Summary (as recorded)

- The 6 gal chili batch read 88°F at the 2 h mark (limit 70°F). It was reheated to 165°F and re-cooled in 2-in pans with ice wands, passing both checks.
- Suspected cause: blast chiller not pulling down; display showed error E4. See ticket TKT-017.

3) Food Safety Limits (cooling and reheat)

- Cooling must meet both limits below. If either fails, reheat to 165°F and restart cooling.
- Reheat of previously cooled food: to 165°F within 2 h, then recool per limits.

Step	Limit	Target	Notes
135°F to 70°F	Within 2 h	≤ 70°F at 2 h	First cooling check
70°F to 41°F	Within 4 h more (total ≤ 6 h)	≤ 41°F by 6 h	Second cooling check
Reheat (if cooling failed)	165°F	Within 2 h	Stir, verify center temp

4) Immediate Containment (what was done)

1. Stopped the chill at the 2 h check when the batch probed 88°F (limit 70°F).
2. Returned chili to stove; reheated rapidly to 165°F, stirring the batch to heat evenly.
3. Divided into shallow 2-in hotel pans, inserted ice wands, and reloaded to blast chiller.
4. Verified both cooling checks passed on the second attempt.
5. Logged readings and opened equipment ticket TKT-017 for error E4.

5) Re-cooling Procedure for Chili (standardize this)

6. After reheat to 165°F, move product to clean, sanitized 2-in pans.

7. Fill pans no deeper than 2 in; leave headspace for airflow.
8. Insert sanitized ice wand in each pan; agitate every 10–15 min to break heat pockets.
9. Place pans in blast chiller with at least 1 in clearance on all sides; do not stack.
10. Set chiller to hard chill program; verify fan and door seal.
11. Probe the coldest-recovering pan:
 - Record at 2 h (target $\leq 70^{\circ}\text{F}$).
 - Record at 6 h total cooling (target $\leq 41^{\circ}\text{F}$).
12. Remove ice wands once product hits $\leq 70^{\circ}\text{F}$ to avoid over-dilution; replace lids and continue to 41°F .
13. Label and move to walk-in. Store above raw and below ready-to-eat garnishes.

6) Monitoring and Logging (how to check and record)

- Use a calibrated digital probe; sanitize between pans.
- Check multiple pans: probe center and thickest area, avoiding wand contact.
- Record actual times and temperatures; note corrective actions if limits missed.

Checkpoint	Time From Start	Target Temp	Record Actual	Initials
Start chill (post-portion)	0 h	135°F or lower entering unit		
First check	2 h	$\leq 70^{\circ}\text{F}$		
Second check	≤ 6 h total	$\leq 41^{\circ}\text{F}$		

- If any pan fails, treat the whole batch as failed for corrective action.

7) Troubleshooting: Blast Chiller Not Pulling Down

If product temp stalls or rises, or display shows error:

- Error observed: E4 on display. Ticket reference: TKT-017.
- Quick checks (do in order):
 1. Verify door fully closed; gasket intact; no gaps.
 2. Reduce load: no stacking; 1 in clearance around pans.
 3. Switch to hard chill mode; confirm fans blowing.
 4. Check that pans are ≤ 2 in deep; redistribute if deeper.
 5. Confirm ice wands are fully frozen and inserted; replace warm wands.
 6. Probe ambient in chiller; if ambient $> 38\text{--}40^{\circ}\text{F}$ after 10 min with door closed, escalate.

- If E4 persists or ambient not dropping:
 - Stop using the unit for active cooling.
 - Move product to ice bath plus shallow pans and ice wands to meet limits.
 - Notify Sous Chef and log details; attach to TKT-017.

8) What To Do If Limits Are Missed (decision tree)

- At 2 h > 70°F (as in this incident: 88°F):
 7. Reheat to 165°F within 2 h.
 8. Re-cool in 2-in pans with ice wands and blast chiller or ice bath.
 9. Recheck at 2 h and 6 h. If both pass, keep.
- At 6 h > 41°F:
 - Discard the batch. Do not attempt to salvage.
- If reheat cannot reach 165°F promptly or quality prevents safe reheat:
 - Discard.

9) Prevention and Station Controls

- Batch size: Limit hot-fill batches for dense items (like chili) to pan depths of 2 in.
- Airflow: Never cover hot pans; vent lids until $\leq 70^{\circ}\text{F}$.
- Load discipline: Stagger loads to avoid overloading the blast chiller.
- Stir/agitate: Use ice wands and stir every 10–15 min until $\leq 70^{\circ}\text{F}$.
- Probe habit: Always set a 2 h timer at start of cooling; set a second timer for 6 h.
- Calibration: Verify thermometers at ice point each shift start.
- Maintenance: Report any alarms, long pull-down times, icing on coils, or door seal issues immediately with a ticket.

10) Roles and Sign-off

- Prep cook: Execute cooling, probing, logging, and first corrective action.
- Samuel Okafor (Sous Chef, Prep): Verify logs, approve corrective actions, and own ticket follow-up (TKT-017).
- Chef on duty: Decide discard vs salvage when reheating is not feasible.